AWS Virtual Private Cloud (VPC) Network Design

Chandupa Silva

May 7, 2026

1 Project Overview

This project demonstrates the design and implementation of a Virtual Private Cloud (VPC) using Amazon Web Services (AWS). The objective was to build a secure and scalable cloud network environment with proper subnet segmentation, routing configuration, and EC2 instance deployment.

2 Objectives

- Design a scalable cloud network using AWS VPC
- Configure public and private subnets for segmentation
- Implement Internet Gateway for external connectivity
- Configure route tables for traffic management
- Deploy and test EC2 instances within the VPC
- Validate network connectivity using real-world testing

3 Network Architecture

The cloud network was designed with the following structure:

- VPC CIDR Block: 15.0.0.0/16
- Public Subnet: 15.0.1.0/24 (for internet-facing resources)
- Private Subnet: 15.0.2.0/24 (for internal resources)
- Internet Gateway attached to the VPC
- Route tables configured for traffic control and internet access

4 Implementation Details

4.1 VPC Creation

A Virtual Private Cloud was created with CIDR block 15.0.0.0/16 to define the overall network boundary.

4.2 Subnet Configuration

Two subnets were created:

- Public Subnet for resources requiring internet access
- Private Subnet for secure internal workloads

4.3 Internet Gateway

An Internet Gateway was attached to the VPC to enable communication between the public subnet and the internet.

4.4 Route Tables

Route tables were configured to:

- Allow internet traffic for public subnet via Internet Gateway
- Restrict direct internet access for private subnet

4.5 EC2 Deployment

An EC2 instance was launched inside the public subnet using a Free Tier eligible configuration. The instance was used to validate connectivity and test network functionality.

5 Testing and Validation

Network testing was performed using:

- EC2 instance connectivity via SSH
- Ping tests to external websites (e.g., google.com)
- Verification of routing and subnet configuration

All tests confirmed successful network configuration and internet connectivity through the VPC.

6 Conclusion

This project successfully demonstrates the implementation of a cloud-based network using AWS VPC. It provides practical experience in subnetting, routing, EC2 deployment, and network troubleshooting, which are essential skills for cloud and network engineering roles.

7 Tools and Services Used

- Amazon Web Services (AWS)
- Virtual Private Cloud (VPC)
- Elastic Compute Cloud (EC2)
- Internet Gateway
- Route Tables
- Linux SSH Terminal